

Charles Zhao

Mechatronics Engineering Student · University of Waterloo

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Personal Website: charlesdehuangzhao.com · PCB · Firmware · Software | [GitHub: BlueCharlie](#)

EDUCATION

University of Waterloo

Sep 2023 – Expected Apr 2028

Honours Mechatronics Engineering, Co-op · Third year

TECHNICAL SKILLS

Electronics: Altium Designer, KiCad, analog front ends, power regulation, PCB assembly and testing

Embedded: C/C++, STC8/8051, ESP32, STM32, Keil C51, UART/SPI/I²C, RS485, Modbus

Software & tools: TypeScript, React, Electron, Node.js, Python, Git, SolidWorks, 3D printing

EXPERIENCE

Minebea Intec GmbH

Jun 2026 – Jul 2026

Embedded Hardware Engineering Intern

- Designed and developed the original PR5170 weighing instrument and its variants across two internships, spanning PCB hardware, embedded firmware, and desktop software.
- Designed or modified more than 20 PCBs across two work periods at Minebea Intec in 2025 and 2026.
- Designed PR5170 standard and Ex ec digital mainboards, plus the Ex ec analog mainboard (gas Zone 2).
- Adapted two Embedded C firmware branches for STC8/8051 acquisition, calibration, filtering, and RS485.
- Developed TConfig in Electron, React, and TypeScript for monitoring, parameter configuration, and calibration across multiple instruments, including PR5170.

Charles Tech

Jul 2025 – Mar 2026

Embedded Systems Engineer & Owner

- Delivered PCB design services for global clients, translating requirements into manufacturing-ready schematics and high-speed layouts.
- Established the engineering workflow and shifted the business toward proprietary electronic product development.

Minebea Intec GmbH

Jan 2025 – Apr 2025

PCB Designer

- Designed the PR5170 standard analog acquisition mainboard and shared power and industrial-interface baseboard, alongside HMI and protection-board variants.
- Integrated ADC and MCU circuitry, regulated supplies, RS232/RS485 interfaces, and 4–20 mA output functions.

Ningbo Chengzi Network Technology Co., Ltd

May 2024 – Aug 2024

Website Developer

- Developed a WeChat Mini Program prototype that attracted more than 100 early users in its first month.
- Improved LLM workflows through prompt engineering and context management; analyzed more than 10 competing products.